

GREAT SCIENTISTS

Albert Einstein

German-born physicist Albert Einstein (1879–1955) was a clerk working in a Swiss patent office when he published four extraordinary scientific papers in 1905. He continued to do groundbreaking work, all of which confirmed him as one of the greatest thinkers of all time and a scientific genius who changed the way we look at the Universe.

Looking at light

In his first paper, Einstein explained the photoelectric effect, a phenomenon in which electrons are sometimes released when light shines on materials. He rejected the idea that light flows as one continuous stream and argued that it was made up of individual parcels of energy known as photons, or “quanta.” His work won him the Nobel Prize in Physics in 1921.

The Special Theory of Relativity

Einstein also showed that the speed of light (186,282 miles/s or 299,792 km/s) is a constant, but time and space are linked and relative. This means that time and space are flexible and can change so that the faster one travels, the slower time passes.

Matter and energy

Einstein altered how science looked at matter and energy in another of his 1905 papers. He introduced his most famous equation: $E = mc^2$ with “E” being energy, “m” being the mass of matter, and “c” being the speed of light. This means that small amounts of matter can contain huge amounts of energy—a principle harnessed both in atomic power and weapons.

Later life

Leaving Europe for the US in 1933, Einstein took a position at Princeton University where he continued his work and enjoyed life as the world’s most brilliant scientist.



“If I were not a physicist, I would probably be a musician. I often think in music. I live my daydreams in music. I see my life in terms of music.”

Albert Einstein,
The Saturday Evening Post, 1929

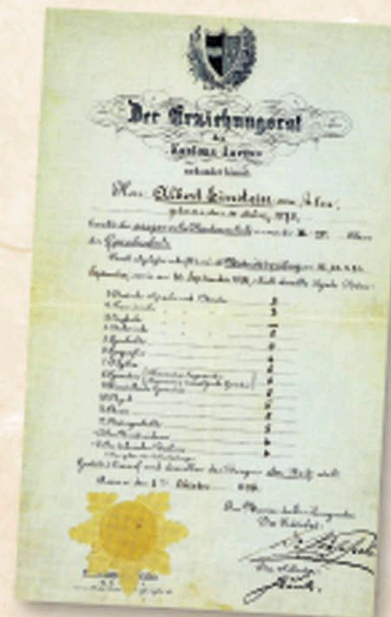
Leisure time

Einstein enjoyed music and took particular pleasure in playing the violin. He also enjoyed sailing, although he never learned to swim.



Young Einstein

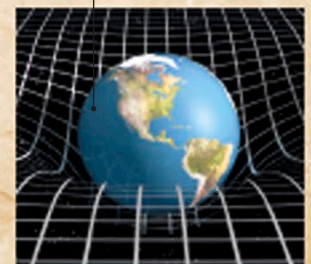
Born to Jewish parents in Ulm, Germany, Einstein (right) had a younger sister named Maja.



School certificate

Einstein’s 1896 Swiss exam certificate gave him the grades to study math and physics at Zürich Polytechnic, Switzerland, at the age of 17. He received top marks in history, physics, algebra, and geometry.

Planet bends space-time (combination of three dimensions of space—length, breadth, and height—with time) creating gravity.



General Theory of Relativity

In 1916, Einstein expanded his special theory of relativity to include gravity.

He showed how objects with large mass bend space-time, causing it to curve like a heavy ball bending a rubber sheet. Smaller objects roll toward the heavy ball due to the curve of the sheet. This theory helped explain black holes and why light from distant stars bends.